

# **100<sub>12</sub> CONCEPTS**

## **THE MATHERA MASTER CONCEPT CODEX**

144 concepts =  $100_{12} = 12^2$  = the 12th Fibonacci number.

A second map beneath the 1,132-skill journey. Skills are practiced in order; Concepts recur across branches and eras. Tiers measure the sophistication of the context in which a Concept has been demonstrated, not the learner's age or school grade.

WORKING ONTOLOGY · OCTOBER 2026

# How to read the Codex

The source curriculum defines seven Eras, 87 Branches and 1,132 Skills, with four practice levels per Skill. This document adds a proposed cross-curricular Concept layer. Era presence and tier ceilings are design mappings inferred from that curriculum, not labels copied verbatim from the source guide.

Universal ceiling: Tier XII. Not every Concept needs twelve meaningful stages. A narrow specialist may top out at Tier V or VII in the current course; a great longitudinal idea can reach Tier XII. That keeps high tiers scarce and informative.

|           |                |
|-----------|----------------|
| COMMON    | up to Tier V   |
| UNCOMMON  | up to Tier VII |
| RARE      | up to Tier IX  |
| EPIC      | up to Tier X   |
| LEGENDARY | up to Tier XI  |
| MYTHIC    | up to Tier XII |

Tier bar: filled pips show the current course ceiling for that Concept. Era chips show where the Concept is expected to receive meaningful practice or application.

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| 43 | Perimeter / Boundary            | COUNT | Tier X   |
| 44 | Congruence                      | COUNT | Tier X   |
| 45 | Symmetry                        | COUNT | Tier XI  |
| 46 | Geometric Transformation        | COUNT | Tier X   |
| 47 | Spatial Reasoning               | COUNT | Tier XI  |
| 48 | Logical Statements              | COUNT | Tier X   |
| 49 | Implication                     | COUNT | Tier X   |
| 50 | Counterexample                  | COUNT | Tier X   |
| 51 | Deduction                       | COUNT | Tier X   |
| 52 | Invariance                      | COUNT | Tier XII |

|     |                                   |         |           |
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| 53  | Generalization                    | COUNT   | Tier X    |
| 54  | Data Collection                   | COUNT   | Tier X    |
| 55  | Approximation by Refinement       | COUNT   | Tier XII  |
| 56  | Accumulation                      | COUNT   | Tier X    |
| 57  | Orthogonality                     | COUNT   | Tier XI   |
| 58  | Multiplication                    | OPERATE | Tier XII  |
| 59  | Division                          | OPERATE | Tier X    |
| 60  | Distributivity                    | OPERATE | Tier XII  |
| 61  | Fractions                         | OPERATE | Tier X    |
| 62  | Ratio                             | OPERATE | Tier X    |
| 63  | Proportion                        | OPERATE | Tier X    |
| 64  | Rate                              | OPERATE | Tier X    |
| 65  | Decimals                          | OPERATE | Tier VIII |
| 66  | Factors & Divisibility            | OPERATE | Tier VII  |
| 67  | Common Factors                    | OPERATE | Tier VIII |
| 68  | Variables                         | OPERATE | Tier X    |
| 69  | Expressions                       | OPERATE | Tier X    |
| 70  | Substitution                      | OPERATE | Tier X    |
| 71  | Simplification                    | OPERATE | Tier X    |
| 72  | Collecting Like Terms             | OPERATE | Tier X    |
| 73  | Expanding Brackets                | OPERATE | Tier XI   |
| 74  | Factoring                         | OPERATE | Tier XI   |
| 75  | Equation Solving                  | OPERATE | Tier XII  |
| 76  | Powers                            | OPERATE | Tier XI   |
| 77  | Mapping                           | OPERATE | Tier X    |
| 78  | Function                          | OPERATE | Tier XII  |
| 79  | Function Representation           | OPERATE | Tier X    |
| 80  | Proportionality                   | OPERATE | Tier X    |
| 81  | Similarity & Scale                | OPERATE | Tier X    |
| 82  | Center / Typical Value            | OPERATE | Tier X    |
| 83  | Rearranging Formulae              | RELATE  | Tier IX   |
| 84  | Inequality Solving                | RELATE  | Tier IX   |
| 85  | Constraints & Solution Sets       | RELATE  | Tier IX   |
| 86  | Signed Quantity                   | RELATE  | Tier IX   |
| 87  | Number Sets                       | RELATE  | Tier VII  |
| 88  | Percent                           | RELATE  | Tier VII  |
| 89  | Roots                             | RELATE  | Tier IX   |
| 90  | Graphing Relations & Functions    | RELATE  | Tier IX   |
| 91  | Intercepts, Zeros & Intersections | RELATE  | Tier IX   |
| 92  | Domain & Range                    | RELATE  | Tier IX   |
| 93  | Linearity                         | RELATE  | Tier IX   |
| 94  | Slope / Gradient                  | RELATE  | Tier X    |
| 95  | Rate of Change                    | RELATE  | Tier XI   |
| 96  | Growth & Decay                    | RELATE  | Tier IX   |
| 97  | Parameters & Families             | RELATE  | Tier IX   |
| 98  | Interpolation & Extrapolation     | RELATE  | Tier IX   |
| 99  | Circular / Angular Measure        | RELATE  | Tier IX   |
| 100 | Sets                              | RELATE  | Tier IX   |
| 101 | Relations                         | RELATE  | Tier IX   |
| 102 | Counting Outcomes                 | RELATE  | Tier VI   |
| 103 | Probability                       | RELATE  | Tier X    |
| 104 | Simulation & Randomness           | RELATE  | Tier IX   |
| 105 | Spread / Variation                | RELATE  | Tier IX   |

|     |                                   |        |           |
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| 106 | Distribution                      | RELATE | Tier IX   |
| 107 | Correlation & Association         | RELATE | Tier IX   |
| 108 | Recurrence                        | SOLVE  | Tier VIII |
| 109 | Series / Summation                | SOLVE  | Tier VIII |
| 110 | Composition                       | SOLVE  | Tier VIII |
| 111 | Inverse Functions                 | SOLVE  | Tier VIII |
| 112 | Function Transformation           | SOLVE  | Tier VIII |
| 113 | Asymptotic Behaviour              | SOLVE  | Tier VIII |
| 114 | Systems / Simultaneity            | SOLVE  | Tier VIII |
| 115 | Necessary & Sufficient Conditions | SOLVE  | Tier VIII |
| 116 | Induction / Recursion             | SOLVE  | Tier VIII |
| 117 | Conditional Probability           | SOLVE  | Tier VI   |
| 118 | Independence                      | SOLVE  | Tier VIII |
| 119 | Expected Value                    | SOLVE  | Tier VI   |
| 120 | Sampling                          | SOLVE  | Tier VIII |
| 121 | Optimization                      | SOLVE  | Tier VIII |
| 122 | Infinite Process & Convergence    | SOLVE  | Tier VIII |
| 123 | Projection                        | PROVE  | Tier VII  |
| 124 | Trigonometric Ratio               | PROVE  | Tier VII  |
| 125 | Geometric Locus                   | PROVE  | Tier VII  |
| 126 | Proof                             | PROVE  | Tier IX   |
| 127 | Statistical Inference             | CHANGE | Tier V    |
| 128 | Limit                             | CHANGE | Tier VI   |
| 129 | Continuity                        | CHANGE | Tier VI   |
| 130 | Derivative                        | CHANGE | Tier VI   |
| 131 | Local Linearization               | CHANGE | Tier VI   |
| 132 | Integral                          | CHANGE | Tier VI   |
| 133 | Fundamental Theorem Connection    | CHANGE | Tier VI   |
| 134 | Differential Equation             | CHANGE | Tier VI   |
| 135 | Series Approximation              | CHANGE | Tier VI   |
| 136 | Vector                            | SPACE  | Tier V    |
| 137 | Scalar & Vector Combination       | SPACE  | Tier V    |
| 138 | Linear Combination & Span         | SPACE  | Tier V    |
| 139 | Linear Independence               | SPACE  | Tier V    |
| 140 | Basis & Dimension                 | SPACE  | Tier V    |
| 141 | Matrix                            | SPACE  | Tier V    |
| 142 | Linear Transformation             | SPACE  | Tier V    |
| 143 | Determinant                       | SPACE  | Tier V    |
| 144 | Eigenstructure                    | SPACE  | Tier V    |

# 001 · Counting

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Reliable enumeration: one object, one count, stable sequence.

Tier logic. Early tiers establish the concrete or procedural form of counting; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 002 · Cardinality

UNCOMMON  
MAX TIER VI



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE

Tracks transferable mastery of cardinality whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of cardinality; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 003 · Number Representation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of number representation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of number representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 004 • Ordering

EPIC  
MAX TIER X



First appears: Era 1 • Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of ordering whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of ordering; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 005 • Place Value

UNCOMMON  
MAX TIER VII



First appears: Era 1 • Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE

Tracks transferable mastery of place value whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of place value; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 006 • Addition

MYTHIC  
MAX TIER XII



First appears: Era 1 • Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of addition whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of addition; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 007 • Subtraction

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of subtraction whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of subtraction; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 008 • Pattern Recognition

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of pattern recognition whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of pattern recognition; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 009 • Classification

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of classification whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of classification; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 010 · Shape

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of shape whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of shape; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 011 · Position

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of position whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of position; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 012 · Distance

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of distance whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of distance; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 013 · Units

LEGENDARY  
MAX TIER XI



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of units whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of units; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 014 · Measurement

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of measurement whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of measurement; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 015 · Data Representation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of data representation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of data representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 016 • Estimation

LEGENDARY  
MAX TIER XI



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of estimation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of estimation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 017 • Repeated Operations / Iteration

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of repeated operations / iteration whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of repeated operations / iteration; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 018 • Algorithms

LEGENDARY  
MAX TIER XI



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of algorithms whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of algorithms; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 019 · Commutativity

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of commutativity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of commutativity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 020 · Associativity

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of associativity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of associativity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 021 · Identity & Inverse Elements

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of identity & inverse elements whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of identity & inverse elements; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 022 · Inverse Operations

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of inverse operations whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of inverse operations; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 023 · Area

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of area whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of area; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 024 · Volume

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of volume whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of volume; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 025 · Precision & Error

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of precision & error whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of precision & error; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 026 · Mental & Computational Fluency

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of mental & computational fluency whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of mental & computational fluency; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 027 · Equality

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of equality whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of equality; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 028 · Equivalence

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of equivalence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of equivalence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 029 · Magnitude

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of magnitude whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of magnitude; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 030 · Scaling

LEGENDARY  
MAX TIER XI



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of scaling whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of scaling; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 031 • Sequence

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of sequence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of sequence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 032 • Coordinates

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of coordinates whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of coordinates; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 033 • Tables

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of tables whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of tables; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 034 · Graphs

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of graphs whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of graphs; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 035 · Changing Representation

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Preserving the mathematical object while moving among words, pictures, tables, symbols, graphs and coordinates.

Tier logic. Early tiers establish the concrete or procedural form of changing representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 036 · Symbolic Representation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of symbolic representation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of symbolic representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 037 · Visual Representation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of visual representation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of visual representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 038 · Reading Graphs

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of reading graphs whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of reading graphs; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 039 · Periodicity

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of periodicity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of periodicity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 040 · Mathematical Modelling

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of mathematical modelling whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of mathematical modelling; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 041 · Angle

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of angle whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of angle; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 042 · Parallel & Perpendicular

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of parallel & perpendicular whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of parallel & perpendicular; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 043 · Perimeter / Boundary

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of perimeter / boundary whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of perimeter / boundary; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 044 · Congruence

EPIC  
MAX TIER X



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of congruence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of congruence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 045 · Symmetry

LEGENDARY  
MAX TIER XI



First appears: Era 1 · Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of symmetry whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of symmetry; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 046 · Geometric Transformation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of geometric transformation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of geometric transformation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 047 · Spatial Reasoning

LEGENDARY  
MAX TIER XI



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of spatial reasoning whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of spatial reasoning; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 048 · Logical Statements

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of logical statements whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of logical statements; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 049 • Implication

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of implication whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of implication; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 050 • Counterexample

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of counterexample whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of counterexample; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 051 • Deduction

EPIC  
MAX TIER X



First appears: Era 1 • Count Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of deduction whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of deduction; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 052 · Invariance

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of invariance whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of invariance; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 053 · Generalization

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of generalization whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of generalization; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 054 · Data Collection

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of data collection whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of data collection; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 055 · Approximation by Refinement

MYTHIC  
MAX TIER XII



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Improving an answer by finer partitions, smaller steps, better local models or repeated iteration.

Tier logic. Early tiers establish the concrete or procedural form of approximation by refinement; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 056 · Accumulation

EPIC  
MAX TIER X



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of accumulation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of accumulation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 057 · Orthogonality

LEGENDARY  
MAX TIER XI



First appears: Era 1 · Count    Course presence: 1:COUNT 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Perpendicularity grown into dot products, projections, least squares, orthogonal bases and Fourier structure.

Tier logic. Early tiers establish the concrete or procedural form of orthogonality; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 058 • Multiplication

MYTHIC  
MAX TIER XII



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Scaling and combining equal groups; later extends to algebraic, functional, vector and matrix products.

Tier logic. Early tiers establish the concrete or procedural form of multiplication; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 059 • Division

EPIC  
MAX TIER X



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of division whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of division; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 060 • Distributivity

MYTHIC  
MAX TIER XII



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

The bridge between additive and multiplicative structure; powers expansion, factoring and much algebra.

Tier logic. Early tiers establish the concrete or procedural form of distributivity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 061 · Fractions

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of fractions whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of fractions; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 062 · Ratio

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of ratio whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of ratio; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 063 · Proportion

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of proportion whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of proportion; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 064 · Rate

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of rate whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of rate; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 065 · Decimals

RARE  
MAX TIER VIII



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 6:CHANGE

Tracks transferable mastery of decimals whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of decimals; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 066 · Factors & Divisibility

UNCOMMON  
MAX TIER VII



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE

Tracks transferable mastery of factors & divisibility whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of factors & divisibility; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 067 · Common Factors

RARE  
MAX TIER VIII



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 6:CHANGE

Tracks transferable mastery of common factors whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of common factors; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 068 · Variables

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of variables whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of variables; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 069 · Expressions

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of expressions whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of expressions; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 070 · Substitution

EPIC  
MAX TIER X



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of substitution whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of substitution; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 071 · Simplification

EPIC  
MAX TIER X



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of simplification whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of simplification; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 072 · Collecting Like Terms

EPIC  
MAX TIER X



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of collecting like terms whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of collecting like terms; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 073 · Expanding Brackets

LEGENDARY  
MAX TIER XI



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of expanding brackets whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of expanding brackets; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 074 · Factoring

LEGENDARY  
MAX TIER XI



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of factoring whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of factoring; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 075 · Equation Solving

MYTHIC  
MAX TIER XII



First appears: Era 2 · Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of equation solving whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of equation solving; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 076 · Powers

LEGENDARY  
MAX TIER XI



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of powers whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of powers; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 077 · Mapping

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of mapping whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of mapping; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 078 · Function

MYTHIC  
MAX TIER XII



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Dependence treated as an object: inputs, outputs, rules, representations, composition, inversion and behavior.

Tier logic. Early tiers establish the concrete or procedural form of function; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 079 · Function Representation

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of function representation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of function representation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 080 · Proportionality

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of proportionality whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of proportionality; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 081 · Similarity & Scale

EPIC  
MAX TIER X



First appears: Era 2 · Operate Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of similarity & scale whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of similarity & scale; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 082 • Center / Typical Value

EPIC  
MAX TIER X



First appears: Era 2 • Operate    Course presence: 2:OPERATE 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of center / typical value whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of center / typical value; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 083 • Rearranging Formulae

RARE  
MAX TIER IX



First appears: Era 3 • Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of rearranging formulae whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of rearranging formulae; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 084 • Inequality Solving

RARE  
MAX TIER IX



First appears: Era 3 • Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of inequality solving whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of inequality solving; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 085 · Constraints & Solution Sets

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of constraints & solution sets whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of constraints & solution sets; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 086 · Signed Quantity

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of signed quantity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of signed quantity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 087 · Number Sets

UNCOMMON  
MAX TIER VII



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 7:SPACE

Tracks transferable mastery of number sets whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of number sets; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 088 · Percent

UNCOMMON  
MAX TIER VII



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 6:CHANGE

Tracks transferable mastery of percent whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of percent; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 089 · Roots

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of roots whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of roots; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 090 · Graphing Relations & Functions

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of graphing relations & functions whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of graphing relations & functions; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 091 · Intercepts, Zeros & Intersections

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of intercepts, zeros & intersections whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of intercepts, zeros & intersections; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 092 · Domain & Range

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of domain & range whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of domain & range; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 093 · Linearity

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of linearity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of linearity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 094 · Slope / Gradient

EPIC  
MAX TIER X



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of slope / gradient whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of slope / gradient; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 095 · Rate of Change

LEGENDARY  
MAX TIER XI



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

How one quantity changes relative to another, from unit rates and slope to derivatives and gradients.

Tier logic. Early tiers establish the concrete or procedural form of rate of change; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 096 · Growth & Decay

RARE  
MAX TIER IX



First appears: Era 3 · Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of growth & decay whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of growth & decay; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 097 · Parameters & Families

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of parameters & families whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of parameters & families; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 098 · Interpolation & Extrapolation

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of interpolation & extrapolation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of interpolation & extrapolation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 099 · Circular / Angular Measure

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of circular / angular measure whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of circular / angular measure; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 100 · Sets

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of sets whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of sets; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 101 · Relations

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of relations whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of relations; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 102 · Counting Outcomes

UNCOMMON  
MAX TIER VI



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE

Tracks transferable mastery of counting outcomes whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of counting outcomes; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 103 · Probability

EPIC  
MAX TIER X



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of probability whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of probability; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 104 · Simulation & Randomness

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of simulation & randomness whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of simulation & randomness; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 105 · Spread / Variation

RARE  
MAX TIER IX



First appears: Era 3 · Relate    Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of spread / variation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of spread / variation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 106 • Distribution

RARE  
MAX TIER IX



First appears: Era 3 • Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of distribution whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of distribution; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 107 • Correlation & Association

RARE  
MAX TIER IX



First appears: Era 3 • Relate Course presence: 3:RELATE 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of correlation & association whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of correlation & association; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 108 • Recurrence

RARE  
MAX TIER VIII



First appears: Era 4 • Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of recurrence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of recurrence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 109 • Series / Summation

RARE  
MAX TIER VIII



First appears: Era 4 • Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of series / summation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of series / summation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 110 • Composition

RARE  
MAX TIER VIII



First appears: Era 4 • Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of composition whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of composition; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 111 • Inverse Functions

RARE  
MAX TIER VIII



First appears: Era 4 • Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of inverse functions whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of inverse functions; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 112 · Function Transformation

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of function transformation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of function transformation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 113 · Asymptotic Behaviour

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of asymptotic behaviour whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of asymptotic behaviour; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 114 · Systems / Simultaneity

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of systems / simultaneity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of systems / simultaneity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 115 · Necessary & Sufficient Conditions

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of necessary & sufficient conditions whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of necessary & sufficient conditions; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 116 · Induction / Recursion

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of induction / recursion whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of induction / recursion; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 117 · Conditional Probability

UNCOMMON  
MAX TIER VI



First appears: Era 4 · Solve Course presence: 4:SOLVE 6:CHANGE

Tracks transferable mastery of conditional probability whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of conditional probability; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 118 · Independence

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of independence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of independence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 119 · Expected Value

UNCOMMON  
MAX TIER VI



First appears: Era 4 · Solve Course presence: 4:SOLVE 6:CHANGE

Tracks transferable mastery of expected value whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of expected value; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 120 · Sampling

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of sampling whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of sampling; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 121 · Optimization

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of optimization whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of optimization; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 122 · Infinite Process & Convergence

RARE  
MAX TIER VIII



First appears: Era 4 · Solve Course presence: 4:SOLVE 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of infinite process & convergence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of infinite process & convergence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 123 · Projection

UNCOMMON  
MAX TIER VII



First appears: Era 5 · Prove Course presence: 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of projection whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of projection; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 124 · Trigonometric Ratio

UNCOMMON  
MAX TIER VII



First appears: Era 5 · Prove Course presence: 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of trigonometric ratio whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of trigonometric ratio; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 125 · Geometric Locus

UNCOMMON  
MAX TIER VII



First appears: Era 5 · Prove Course presence: 5:PROVE 6:CHANGE 7:SPACE

Tracks transferable mastery of geometric locus whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of geometric locus; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

# 126 · Proof

RARE  
MAX TIER IX



First appears: Era 5 · Prove Course presence: 5:PROVE 6:CHANGE 7:SPACE

Why a claim must be true: definitions, implication, counterexample, deduction, construction and general argument.

Tier logic. Early tiers establish the concrete or procedural form of proof; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 127 · Statistical Inference

COMMON  
MAX TIER V



First appears: Era 6 · Change Course presence: 6:CHANGE

Tracks transferable mastery of statistical inference whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of statistical inference; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 128 · Limit

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of limit whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of limit; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 129 · Continuity

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of continuity whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of continuity; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 130 · Derivative

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of derivative whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of derivative; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 131 · Local Linearization

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of local linearization whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of local linearization; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 132 · Integral

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of integral whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of integral; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 133 · Fundamental Theorem Connection

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of fundamental theorem connection whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of fundamental theorem connection; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 134 · Differential Equation

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of differential equation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of differential equation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 135 · Series Approximation

UNCOMMON  
MAX TIER VI



First appears: Era 6 · Change Course presence: 6:CHANGE 7:SPACE

Tracks transferable mastery of series approximation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of series approximation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 136 · Vector

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of vector whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of vector; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 137 · Scalar & Vector Combination

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of scalar & vector combination whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of scalar & vector combination; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 138 · Linear Combination & Span

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of linear combination & span whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of linear combination & span; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 139 · Linear Independence

COMMON  
MAX TIER V



First appears: Era 7 · Space    Course presence: [7:SPACE](#)

Tracks transferable mastery of linear independence whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of linear independence; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 140 · Basis & Dimension

COMMON  
MAX TIER V



First appears: Era 7 · Space    Course presence: [7:SPACE](#)

Tracks transferable mastery of basis & dimension whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of basis & dimension; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 141 · Matrix

COMMON  
MAX TIER V



First appears: Era 7 · Space    Course presence: [7:SPACE](#)

Tracks transferable mastery of matrix whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of matrix; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 142 · Linear Transformation

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of linear transformation whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of linear transformation; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 143 · Determinant

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of determinant whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of determinant; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## 144 · Eigenstructure

COMMON  
MAX TIER V



First appears: Era 7 · Space Course presence: [7:SPACE](#)

Tracks transferable mastery of eigenstructure whenever it reappears in a new representation, application, or level of abstraction.

Tier logic. Early tiers establish the concrete or procedural form of eigenstructure; middle tiers demand symbolic fluency and transfer; upper tiers count only when the same idea survives inside substantially more abstract mathematics.

## Design notes

Why variable tier ceilings? A universal Tier XII gives the system a stable RPG scale, while variable ceilings prevent fake complexity. A Concept only earns a new tier when the object, representation, abstraction, or transfer demand changes materially.

Why keep repeated Skills? Repetition is deliberate. The Skill Tree supplies encounters and spaced practice; the Concept Codex aggregates evidence across those encounters. A learner can therefore revisit a familiar procedure quickly while still strengthening a deep Concept over years.

Why 144? It is not only a joke: 144 sits comfortably inside the desired ontology size. The joke is unusually good, though:  $144_{10} = 100_{12}$ ,  $144 = 12^2$ , and 144 is the 12th Fibonacci number.

Status. This is a first-pass ontology. The next rigorous step is to tag all 1,132 Skills with 1-3 primary Concepts plus secondary/incidental Concepts, then split overloaded Concepts and merge Concepts that receive indistinguishable evidence.